

Business Operations & KPI Governance Case

Power BI · SQL · KPI Logic · Data Quality · Business Analysis

Prepared by Serkan Akdemir

10.64m €	9.75m €	-896.81k €	14.81%	14.32%
Gross Revenue	Net Revenue	Adjustment Revenue	Cancellation Rate	Orders without CustomerID

Executive Summary

This skill dossier demonstrates how raw online retail transaction data can be cleaned, modeled, visualized and validated to support business decisions. The analysis focuses on revenue quality, customer value, cancellation effects and data quality. It combines a star-schema-oriented Power BI model, KPI definitions, three dashboard pages and SQL validation in SQLite.

Takeaway: Revenue volume alone is not sufficient for operational steering. KPI governance must separate gross activity from net revenue quality, cancellation value and customer-identification coverage.

Business Question, Dataset & Analytical Approach

Business question	Why do revenue quality, customer value and operational efficiency develop differently although business activity is visible in the transaction data?
Dataset	UCI Online Retail transaction data with invoice numbers, stock codes, product descriptions, quantities, invoice dates, unit prices, customer IDs and countries.
Analytical focus	Revenue quality, customer value, cancellation effects, country concentration, product/revenue-line risk and CustomerID completeness.
Approach	Prepare raw transactions, define KPI logic, build a star-schema-oriented Power BI model, visualize the findings and validate selected metrics with SQL in SQLite.

Project step	Output	Business purpose
1. Data profiling	Missing IDs, negative quantities, cancellation patterns	Surface data limitations before KPI interpretation
2. Data preparation	Revenue, TransactionType, HasCustomerID, YearMonth	Create auditable analysis logic
3. Semantic model	Fact table plus dimensions	Separate transactions from analysis attributes
4. Dashboarding	3 Power BI pages	Make key findings visible for stakeholders
5. SQL validation	Independent SQL checks	Confirm dashboard logic outside the visualization layer

Takeaway: The project turns transaction activity into decision-oriented KPIs that distinguish commercial performance from operational and data-quality effects.

Data Quality & Star-Schema-Oriented Data Model

Preparation logic

Logic	Business purpose
Revenue = Quantity x UnitPrice	Creates a transaction-level revenue measure for aggregation.
Cancellation logic	Separates regular sales from cancellations and adjustment effects.
HasCustomerID flag	Enables reliable customer-level analysis only where customer identification exists.
YearMonth logic	Supports chronological monthly trend analysis.

Semantic model

FactTransactions / Online Retail is the fact table. DimCalendar, DimCountry, DimCustomer and DimProduct are supporting dimensions for filtering and grouping.



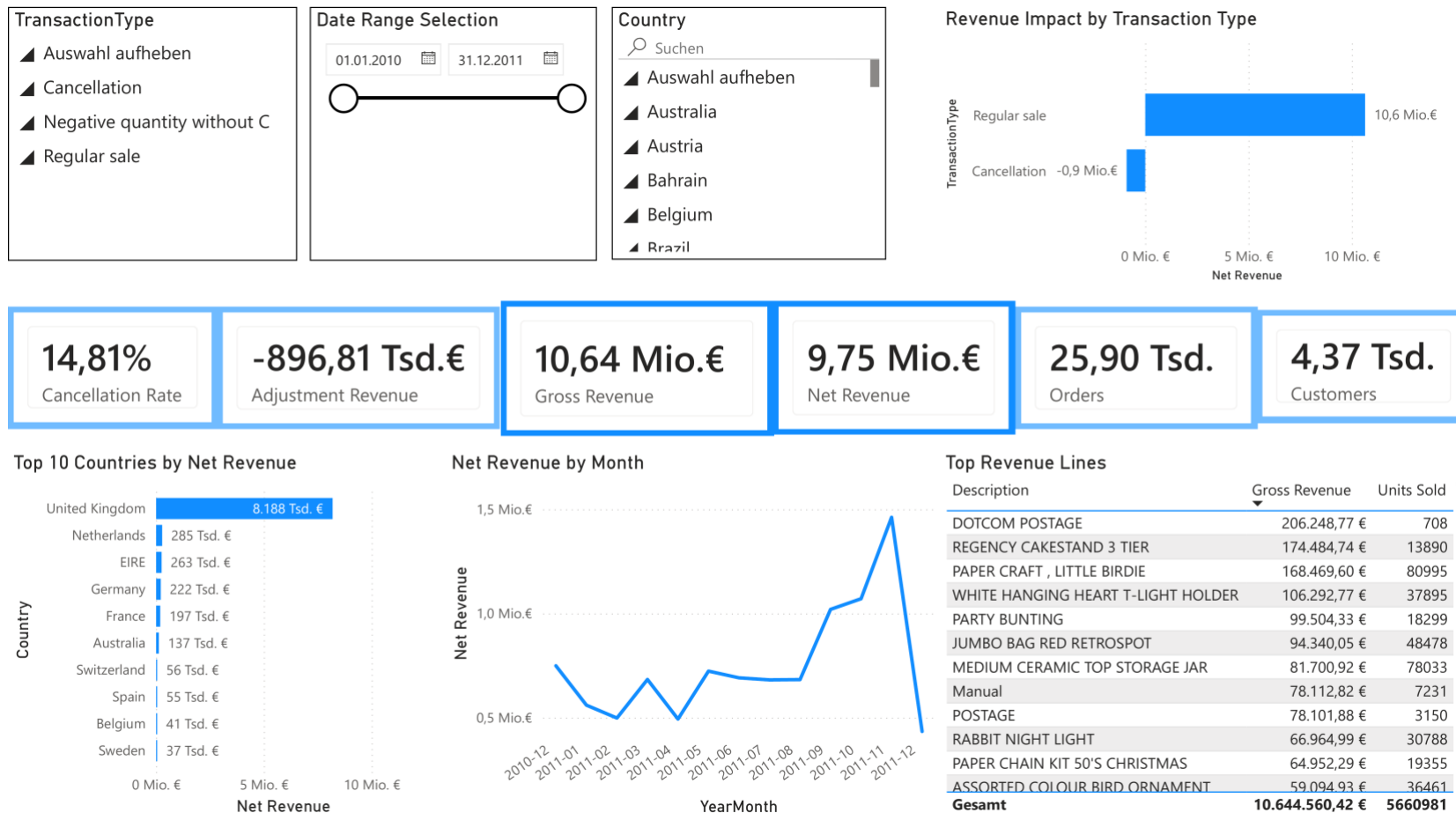
Takeaway: Clear data preparation and model structure make the dashboard easier to validate, maintain and explain to business stakeholders.

KPI Definitions

KPI	Definition	Purpose
Gross Revenue	Revenue before cancellation and adjustment effects.	Measures visible commercial activity.
Net Revenue	Final revenue after negative transactions and cancellations are included.	Measures revenue quality and realized value.
Adjustment Revenue	Difference between gross and net revenue driven by negative transactions.	Highlights revenue erosion.
Cancellation Rate	Cancellation orders divided by total orders.	Measures operational cancellation intensity.
Cancellation Value	Absolute revenue value of cancellation transactions.	Quantifies financial impact of cancellations and reversals.
Known Customer Revenue	Revenue from transactions with CustomerID.	Supports customer-value analysis.
Revenue per Customer	Known customer revenue divided by distinct known customers.	Shows average customer value.
Repeat Customer Rate	Share of known customers with more than one order.	Indicates customer retention pattern.
Orders without CustomerID Rate	Orders without CustomerID divided by total orders.	Measures customer-data completeness risk.

Takeaway: The KPI dictionary makes the report auditable by connecting each number to a definition and management purpose.

Executive Overview + Key Findings

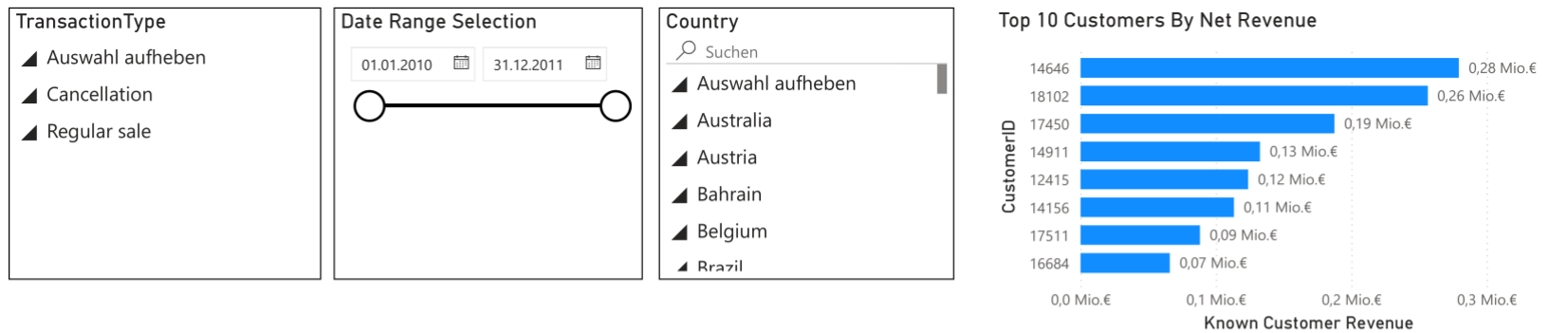


Key Findings

- Gross Revenue is approx. 10.64m €, while Net Revenue is approx. 9.75m €.
- Adjustment Revenue of approx. -896.81k € shows material revenue reduction from cancellations and negative transactions.
- United Kingdom dominates country-level Net Revenue with approx. 8.19m €.

Takeaway: High business activity is visible, but revenue quality is affected by cancellations, negative transactions and strong country concentration.

Customer & Revenue Deep Dive + Key Findings


4,37 Tsd.

Customers

8,30 Mio.€

Known Customer Revenue

1,90 Tsd.€

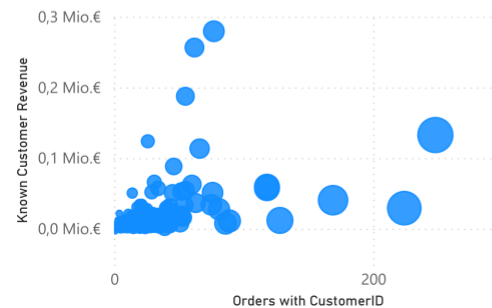
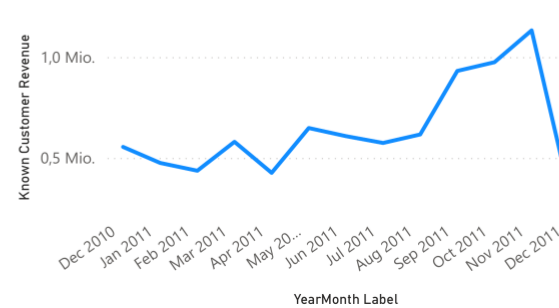
Revenue per Customer

5,08

Orders per Customer

69,97%

Repeat Customer Rate

Customer Revenue Vs Order Count

Known Customer Revenue By Month

Top Customer Table

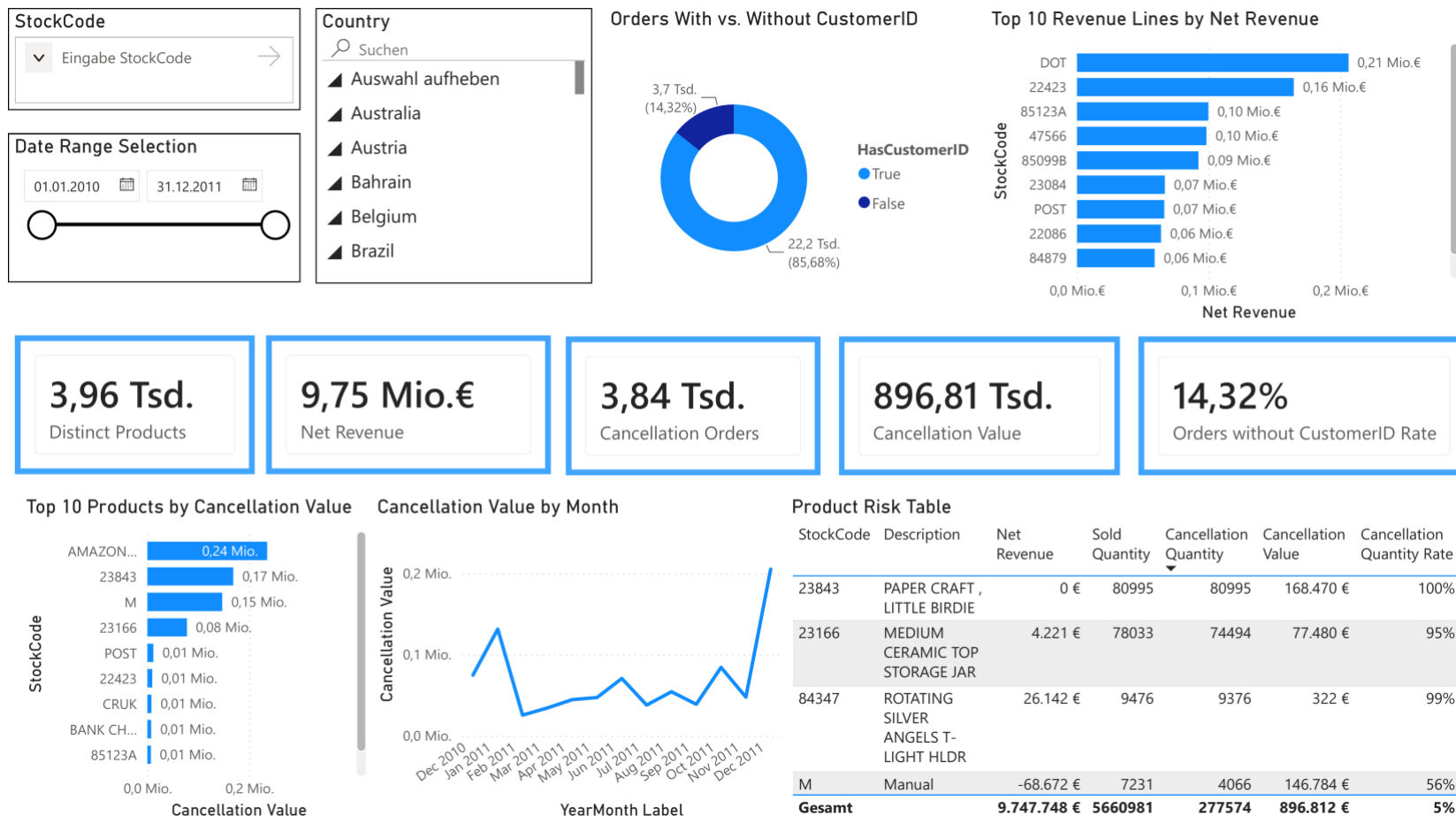
Customer ID	Known Customer Revenue	Orders with CustomerID	Cancellation Orders	Cancellation Rate
14646	279.489 €	77	3	3,9%
18102	256.438 €	62	2	3,2%
17450	187.482 €	55	9	16,4%
14911	132.573 €	248	47	19,0%
12415	123.725 €	26	5	19,2%
14156	113.384 €	66	11	16,7%
17511	88.125 €	46	15	32,6%
Gesamt	8.300.066 €	22190	3654	16,5%

Key Findings

- Known Customer Revenue is approx. 8.30m €; Revenue per Customer is approx. 1.90k €.
- Orders per Customer are 5.08 and Repeat Customer Rate is approx. 69.97%.
- Customer value is concentrated among a limited number of high-value customers.

Takeaway: Known customer revenue is commercially valuable but concentrated; customer analysis depends strongly on CustomerID completeness.

Product & Cancellation Risk + Key Findings



Key Findings

- Distinct Products are approx. 3.96k; Cancellation Orders are approx. 3.84k.
- Cancellation Value is approx. 896.81k € and Orders without CustomerID Rate is approx. 14.32%.
- Some revenue lines show high cancellation values despite low or neutral net revenue.

Takeaway: Cancellation risk is not evenly distributed; product and revenue-line details help identify where business validation is required.

SQL Validation

Validation areas

Example SQL validation excerpts

SQL validation area	Validation purpose
Country Performance	Reproduces country-level net revenue and confirms United Kingdom dominance.
Customer Value	Ranks known customers by revenue and validates customer concentration.
Product Cancellation Risk	Checks cancellation value and identifies revenue lines affected by reversals or negative quantities.
CustomerID Data Quality	Quantifies orders without CustomerID and validates customer-analysis limitations.

```
SELECT Country,
       ROUND(SUM(Revenue), 2) AS net_revenue,
       COUNT(DISTINCT InvoiceNo) AS orders
FROM online_retail_working
GROUP BY Country
ORDER BY net_revenue DESC
LIMIT 10;
```

SQL validation note: Selected dashboard metrics were reproduced with SQL in SQLite directly on transaction-level data. This provides an independent check that Power BI measures align with the prepared source data.

```
SELECT CustomerID,
       ROUND(SUM(Revenue), 2) AS known_customer_revenue,
       COUNT(DISTINCT InvoiceNo) AS orders
FROM online_retail_working
WHERE HasCustomerID = 1
GROUP BY CustomerID
ORDER BY known_customer_revenue DESC
LIMIT 10;
```

Takeaway: SQL checks strengthen KPI trust by validating dashboard results outside the visualization layer.

Recommendations, Limitations & Next Steps

Recommendation		Business rationale	
Establish Cancellation Value as an operational KPI.		Shows whether revenue growth is stable or reduced by reversals and corrections.	
Review product and revenue lines with high cancellation value.		Identifies potential process issues, special bookings or distorted product performance.	
Improve CustomerID data quality.		Strengthens customer value, repeat purchase and segmentation analysis.	
Monitor top customers by revenue, order frequency and cancellation behavior.		Separates stable high-value customers from risk-prone customer relationships.	
Use Power BI dashboard plus SQL checks as a KPI governance process.		Keeps definitions, calculations and results reproducible.	
Limitations		Next steps	
Public dataset without full business context; December 2011 is not a full month; missing CustomerIDs limit customer-level analysis; not every revenue line is a classic product; the dashboard identifies patterns and hypotheses, not final root causes.		Validate top cancellation lines with business stakeholders; finalize a KPI dictionary; monitor CustomerID coverage monthly; keep SQL validation as a repeatable control; extend the model with reason codes, channels or product categories if available.	
Takeaway: The dossier shows a complete analyst workflow: data preparation, model structure, KPI logic, dashboard communication, SQL validation and business recommendations.			